

Roll No.

DEMM-11

B. TECH. (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

PRESSURE DIE CASTING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define the purpose of sprue, runner and riser in casting. A sprue in a sand mold has a top diameter of 25 mm and height of 250 mm, the velocity of the molten metal at the entry of the sprue is 1.5 m/s, assume acceleration due to gravity as 9.8 m/s^2 and neglect all losses. If the mould is well ventilated, find out the velocity of the molten metal at the bottom of sprue. (CO1)

OR

- (b) Compare and contrast Gravity Die Casting and Pressure Die Casting with their applications, advantages, and limitations. (CO2)

P. T. O.

2. (a) Explain various casting defects inspection methods used in the casting industry. (CO1)

OR

- (b) Calculate the volume of the cylindrical riser and solidification time for the cylindrical riser using the following data : (CO1)

Diameter of the cylindrical riser (D) = 20 cm, Height of the cylindrical riser (H) = 30 cm, Thermal conductivity of the casting material (k) = 0.8 cm/sec, Latent heat of fusion (L) = 300 J/g, Density of the casting material (ρ) = 2.5 g/cm³, Ambient temperature (T_a) = 25°C, Initial casting temperature (T_i) = 700°C.

3. (a) Explain the Time temperature transformation (TTT) diagram showing the various regions for Steel. (CO1/CO2)

OR

- (b) Discuss the metallurgy of aluminium alloys, focusing on alloying elements and their properties. Provide examples of how different alloying elements influence the characteristics and performance of aluminium alloys. (CO1/CO2)

4. (a) Discussed the casting defects that occur in the manufacturing process, their root causes and suggest effective remedies to prevent them.

(CO1/CO2)

OR

- (b) Discuss the different surface hardening techniques to impart hardness on cast components in detail. (CO1/CO2)

(3)

5. (a) Explain the various heat treatment processes and effect of cooling rate on microstructure of manufactured component. (CO1/CO2)

OR

- (b) Explain the function of Chills, Padding, Core, Core prints, Fluxes in Sand casting. (CO1/CO2)

Roll No.

TME-606

B. TECH. (ME) (SIXTH SEMESTER) MID SEMESTER EXAMINATION, March, 2024

CAD/CAM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define a typical product cycle with flowchart and suitable example.

(CO1)

OR

- (b) What does CAD stand for ? Describe the significance of CAD in both manufacturing and design engineering.

(CO1)

2. (a) Enumerate the benefits of Computer-Aided Manufacturing (CAM) compared to traditional manufacturing methods, detailing them in ten points.

(CO1)

OR

- (b) Enumerate the constraints associated with NC (Numerical Control) and CNC (Computer Numerical Control) machine tools.

(CO1)

P. T. O.

(2)

3. (a) Explain the difference between Direct Numerical Control and Distributed Numerical Control. (CO2)

OR

- (b) What are the advantages of DNC and DNC communication ? (CO2)

4. (a) What do you mean by Network Topology ? (CO2)

OR

- (b) Discuss in detail about Bus, Ring, Star, Mesh and Tree topology. (CO2)

5. (a) Define the concept of Group Technology and elaborate on the process of establishing machine cells through the application of Group Technology. (CO3)

OR

- (b) What do you mean by CAPP ? What are its benefits ? (CO3)

Roll No.

TME-602

B. TECH. (ME) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024
DESIGN OF MACHINE ELEMENTS-II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

(iii) Assume suitable data if required.

1. (a) Classify springs according to their shapes. Draw neat sketches indicating in each case whether stresses are induced by bending or by torsion Explain the following terms of the spring : (CO1)

(i) Free length

(ii) Solid height

(iii) Spring rate

(iv) Active and inactive coils

(v) Spring index; and

(vi) Stress factor.

P. T. O.

- (b) List the materials commonly used for the manufacture of the leaf springs and write the expression for determining the stress and deflection in full length and graduated leaves. (CO1)
2. (a) A safety valve, 40 mm in diameter, is to blow off at a pressure of 1.2 MPa. It is held on its seat by means of a helical compression spring, with initial compression of 20 mm. The maximum lift of the valve is 12 mm. The spring index is 6. The spring is made of cold drawn steel wire with ultimate tensile strength of 1400 N/mm². The permissible shear stress can be taken as 50% of this strength. ($G = 81370 \text{ N/mm}^2$).
- Calculate : (CO1/CO2)
- (i) wire diameter
 - (ii) mean coil diameter; and
 - (iii) number of active coils.
- (b) (i) What are the advantages of V-belts over flat belts ? Why are belt drives called 'flexible' drives ?
- (ii) State the advantages/disadvantages and applications of chain drives. (CO1/CO2)
3. (a) A semi-elliptic leaf spring used for automobile suspension consists of three extra full-length leaves and 15 graduated-length leaves, including the master leaf. The centre-to-centre distance between two eyes of the spring is 1 m. The maximum force that can act on the spring is 75 kN.

For each leaf, the ratio of width to thickness is 9 : 1. The modulus of elasticity of the leaf material is 207000 N/mm^2 . The leaves are pre-stressed in such a way that when the force is maximum, the stresses induced in all leaves are same and equal to 450 N/mm^2 .

Determine : (i) the width and thickness of the leaves; (ii) the initial nip; and (iii) the initial pre-load required to close the gap C between extra full-length leaves and graduated-length leaves. (CO1, CO2)

- (b) A simple roller chain 10B is used to drive the camshaft of an internal combustion engine. Both shafts rotate at 350 rpm and the centre distance between their axes should be approximately 550 mm. The number of teeth on each sprocket wheel is 19.

Calculate : (i) the number of chain links; and (ii) the correct centre distance. (CO2/CO2)

4. (a) A helical valve spring is to be designed for an operating load range of approximately 90 to 135 N. The deflection of the spring for the load range is 7.5 mm. Assume a spring index of 10. Permissible shear stress for the material of the spring = 480 MPa and its modulus of rigidity = 80 kN/mm^2 . Design the spring. (CO1/CO2)

- (b) Sketch the cross section of a V-belt and label its important parts with brief explanation. How will you designate V-belt ? State advantages and disadvantages of V-belt drive over flat belt drive. (CO1/CO2)

5. (a) A V-belt drive consists of three V-belts in parallel on grooved pulleys of the same size. The angle of groove is 30° and the coefficient of friction 0.12. The cross-sectional area of each belt is 800 mm^2 and the permissible safe stress in the material is 3 MPa. Calculate the power that can be transmitted between two pulleys 400 mm in diameter rotating at 960 r.p.m. (CO2)
- (b) Give the application of the following wire ropes : (CO2)
- (i) 6×7 rope
 - (ii) 6×19 rope
 - (iii) 6×37 rope.

Under what circumstances a fibre rope and a wire rope is used ? What are the advantages of a wire rope over fibre rope ?

Roll No.

TME-609

B. TECH. (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

INDUSTRIAL ENGINEERING AND PROJECT MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

(iii) Draw neat diagrams as necessary.

(iv) Make assumptions for any missing data.

1. (a) Explain the key steps involved in the layout design process and discuss the importance of each step in optimizing plant layout. (CO1)

OR

- (b) Analyze the various factors affecting the productivity of materials, buildings, machines, and power within a specific industrial setting (use manufacturing example). Recommend strategies for improvement in each area. (CO1)

P. T. O.

2. (a) Discuss the role of human factors in work study, considering aspects like fatigue, ergonomics, and motivation. Analyze how these factors can impact work efficiency and suggest work study techniques to address them. (CO1)

OR

- (b) A study finds that implementing hand tools with better grips could reduce worker fatigue and potentially increase production by 2%. If the current production rate is 100 units per hour, calculate the potential increase in production per hour. Briefly explain one additional benefit of improving ergonomics in the workplace besides productivity. (CO1)
3. (a) Explore the relationship between work study and various stakeholders, including management, supervision, and workers. Explain how effective work study practices benefit each group and contribute to improved collaboration and performance. (CO1)

OR

- (b) Discuss the core principles of work design and explain how they can contribute to improved worker performance, satisfaction, and overall productivity. Provide real-world examples to illustrate your points. (CO2)
4. (a) Compare and contrast two different method analysis techniques (e.g., flow process chart, work sampling) highlighting their strengths, weaknesses, and suitability for different types of tasks or processes. (CO2)

(3)

OR

- (b) Choose a specific time study technique (e.g., stopwatch time study, predetermined motion time systems) and explain its key features, advantages, and limitations. Discuss the importance of accuracy and reliability in time study data collection. (CO2)
3. (a) Explain the key rating factors used in time studies and how they influence the determination of standard times. Discuss the potential biases that may arise during rating and suggest strategies to minimize subjective judgment. (CO2)

OR

- (b) A worker completes a task in 25 seconds, which is rated as 15% faster than the average worker. If the standard time for the task is 30 seconds, calculate the adjusted task time based on the performance rating. Discuss the potential impact of subjective rating on standard time accuracy. (CO2)



Roll No.

DETF-03

B. TECH. (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

REFRIGERATION AND AIR CONDITIONING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Differentiate heat engine, refrigerator, and a heat pump. (CO3)

OR

- (b) A machine working on a Carnot cycle operates between 310 K and 270 K. Calculate the COP when it is operated as : a heat pump, heat engine and a refrigeration machine. (CO3)

2. (a) Derive the formula for COP of reverse Brayton cycle. (CO2)

OR

- (b) A refrigerator working on Bell-Coleman cycle operates b/w pressure limits of 1.05 bar and 8.5 bar. Air is drawn from the cold chamber at 10° C, compressed and then it is cooled to 30° C before entering to the expansion cylinder. The expansion and compression follows the law $PV^{1.3} = C$. Determine the theoretical COP of the system. (CO2)

P. T. O.

3. (a) Classify air refrigeration systems used for aircraft and explain bootstrap air refrigeration systems in details. (CO1)

OR

- (b) A simple evaporative air refrigeration system is used to take 25 tonnes of refrigeration load. The ambient air conditions are 23°C and 0.85 bar. The ambient air is rammed isentropically to a pressure of 1.01 bar. The air leaving the main compressor at a pressure of 4 bar is first cooled in the heat exchanger having effectiveness of 0.65 and then in the evaporator where its temperature is reduced by 5°C . The air from the evaporator is passed through the cooling turbine and then supplied to the cabin. The cabin is to be maintained at 25°C and 1.05 bar. If the internal efficiency of the compressor is 80% and that of cooling turbine 75%, determine :

- (i) mass of air bled off from the main compressor,
(ii) power required for the refrigerating system, and (iii) COP of the refrigerating system. (CO1)

4. (a) What are the factors, affect the performance of simple vapor compression system. Show the effect of (i) discharge pressure increase and (ii) subcooling with the help of P-h-diagram and calculation. (CO1)

OR

- (b) Explain actual vapour compression refrigeration cycle with T-S diagram. (CO1)

(3)

5. (a) What is the need of multi-compression refrigeration system. (CO2)

OR

- (b) Draw a schematic for multi evaporator system with a single compressor and multiple expansion valves and explain the process with calculation. (CO2)

